Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/28/2021
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Effects of Modeled Microgravity on the Induction of Bacteria-Induced Apoptosis During Animal Development		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 9/15/21)		
Key Personnel Changes/Previous PI:	April 2021 report: No changes		
COI Name (Institution):			
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Task Description:	Long-duration space flight imposes numerous physiological challenges to the human body, such as bone loss, dysregulation of the immune system, and radiation exposure. Although considerable progress has been made in delineating these biological phenomena, one area of research that has received little attention is the impact that space flight has on those commensal, non-pathogenic microbes that associate with human tissues. To address this gap in knowledge we propose to examine how microgravity alters normal bacteria-induced development in animal tissues. Specifically, we propose to elucidate the pathways associated with bacteria-induced apoptosis, a type of programmed cell death that is critical in maintaining tissue homeostasis and the immune response of animals. To address this goal, the monospecific symbiosis between the bobtail squid Euprymna scolopes and its luminescent partner Vibrio fischeri will be used as a model system to examine the effects of modeled microgravity (i.e., High Aspect Ratio Vessel) on bacteria-induced apoptosis in the host animal. The central hypothesis of this project is that microgravity alters the extrinsic and intrinsic signaling pathways associated with bacteria-induced apoptosis in the host animal. Our previous studies have shown that modeled microgravity causes an acceleration of bacteria-induced apoptosis in the host; however, the mechanisms underlying the developmental acceleration is unknown. To address this issue we propose the following specific objectives: Objective 1. Delineate the impact of modeled microgravity on the shedding of apoptosis-inducing lipopolysaccharide by the bacterial symbiont. We hypothesize that the acceleration of cell death in the host tissues is due to the increased shedding of bacterial lipopolysaccharide (LPS), thereby triggering an upregulation of genes associated with the extrinsic apoptosis pathway. To address this objective we will use a three-pronged approach to examine the controls on LPS-shedding and identify the host molecular response to the shedding using NanoString, western blotting, and immunocytochemistry under modeled microgravity conditions. Objective 2. Elucidate changes in activation and mediation of the extrinsic and intrinsic apoptosis pathways associated with bacteria-induced apoptosis in microgravity. By data mining currently available transcriptomes, we will monitor the differential expression of potential target ligands, receptors, adaptor proteins, and inhibitors of apoptosis. Objective 3. Examine the onset of caspase activity during accelerated apoptosis in host light organ under simulated microgravity conditions and attempt to modulate its onset. We will examine the differential expression of caspases within the light organ under modeled microgravity conditions and anticipate that the caspases are highly integrated with the innate immune response of the host. Additionally, we will attempt to modulate the protease activity using a range of inhibitors to potentially mitigate the microgravity-induced effects on bacteria-induced apoptosis in the host tissues. Significance of Project - The proposed work has several implications for the field of space biology and directly relates to NASA Space Biology Science Plan and Decadal Survey, specifically the Microbiology (MB) and Cell Molecular Biology (CMB) program elements. First, by examining how the normal bacteria-induced development of animals is mediated under microgravity conditions, we can begin to determine the sensitivity of these associations to perturbations in the space environment. Secondly, by removing gravity as a constant we can address the question of whether gravity obscures aspects of bacteria-induced animal development that may otherwise go undetected under terrestrial conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work has several implications for the field of space biology and life on Earth. First, by examining how healthy animal-microbe interactions are initiated and maintained under microgravity conditions, we can begin to determine the sensitivity of these associations to perturbations in the space environment. There is a gap in our understanding of the effects that the space environment has on the microbiome of animals. As there are more than 2000 identified bacterial species that form associations with humans, there are fewer than 100 species identified as human pathogens, yet most microgravity studies have focused on pathogens. This leaves a large array of undescribed commensal organisms that form interactions with animal cells, which may, or may not, be impacted by microgravity conditions. Here in this proposal, we ask the question of how does microgravity alter key bacteria-induced developmental events within the host animal? In other words, what happens to mutualistic associations when gravity is removed? Secondly, by removing gravity as a constant we can address the question of whether gravity obscures aspects or cues of bacteria-induced animal development that may otherwise go undetected under terrestrial conditions. These new insights from simplified animal model systems can then be applied to our understanding of human physiology in the space environment.
	Summary of Research Progress in Year 3: In this year, we have one manuscript in review, one additional manuscript in preparation, three oral and two poster presentations on research derived from this project. These include: Objective 1. Delineate the impact of modeled microgravity on the shedding of apoptosis-inducing lipopolysaccharide (LPS) by the bacterial symbiont. Most of the activities associated with this objective were completed in YR 1 and the manuscript was published in YR2. We showed that modeled microgravity caused the following: 1. There was a significant increase in LPS shedding by the beneficial V. fischeri under modeled microgravity. 2. The cell membrane of V. fischeri was negatively impacted under modeled microgravity such that a statistically higher number of outer membrane vesicles were released under modeled microgravity conditions 3. The release of the outer membrane vesicles in V. fischeri is tied to motility but even in motility mutants V. fischeri released a higher number of LPS-containing outer membrane vesicles under modeled microgravity conditions Implications: The implications of this work suggest that microgravity may be causing the increased shedding of bacterial endotoxin and, depending on the bacterium, this could be problematic for host organisms. The results of this work have been published in npj Microgravity in 2021. Citation and link to PDF listed in the below Bibliography section.

Task Progress:	Objective 2. Elucidate changes in activation and mediation of the extrinsic and intrinsic apoptosis pathways associated with bacteria-induced apoptosis in microgravity. In previous results we have shown that under modeled microgravity conditions there is an acceleration of the bacteria-induced apoptosis response In YR2 we conducted an experiment where we exposed host animals to ow-shear modeled microgravity (LSMMG) conditions in the presence and absence of symbiotic bacteria. We then isolated the RNA and used the sequencing platform NanoString to examine the expression of specific genes associated with extrinsic and intrinsic apoptosis (i.e., cell death). Using this expression data along with the reference transcriptome we published in 2019 we have now developed a pathway map of many of the apoptotic genes associated with the bacteria induced apoptosis pathway and have ascertained the following: 1. We have identified genes associated with both extrinsic and intrinsic apoptosis. Most of the genes (55%) discovered have a positive effect on apoptosis (i.e., induce apoptosis) whereas 34% of the recovered genes had a negative effect (i.e., suppress apoptosis). Eleven percent of the genes had multiple or undetermined function. Interestingly, the majority of the genes differently expressed during modeled microgravity were associated with the intrinsic apoptosis pathway (40%) suggesting that internal stressors may be triggering the apoptosis pathway in the host animals, whereas only 16% of the observed genes were associated with extrinsic, or external, activation of apoptosis. 2. The highest differentially regulated gene under modeled microgravity is the LPS-binding proteins (LBP1 and LBP3), which recognize the LPS released from bacteria. As we previously showed in Objective 1, the excess LPS released under modeled microgravity may be triggering the accelerated apoptosis and it may be working via the LBP death receptor binding mechanisms. Implications: The results have enabled us to map the apoptosis genetic pathway in the host animal, which has not previously been done, and ascertain the effect of microgravity on the gene expression. We anticipate a paper on these results in the upcoming year. We are attempting to more fully characterize these genes that are differentially expressed in the modeled microgravity. Objective 3. Examine the onset of caspase activity during accelerated apoptosis in host light organ under simulated microgravity conditions and attempt to modulate its onset. 1. One of the major outcomes of Objective 2 was the discovery of numerous caspase-encoding genes for enzymes that are involved in the apoptotic cell death process. We exampled the differential expression of these genes and found that that several of the genes (cas9, cas3, cas8, cas2, and cas10) are all differentially upregulated under modeled microgravity conditions when compare to gravity controls. 2. We are now in the process of evaluating the enzyme activity to ensure that the protein is active during similar time points. The delay due to COVID-19 slowed our progress in this area, but we have now optimized the enzyme detection assays and will be examining caspase activity for additional time points and genes. We will be testing the activity of Caspase 3/7, 8, and 9 enzymes over the first 20 hours post-infection with the bacteria. 3. Lastly, we are trying to assess whether we can mitigate the increase in apoptotic cell death and further elucidate the different mechanisms of caspase activation. We have started to ascertain and optimize the use of inhibitors to prevent the onset of apoptosis. Preliminary data has identified that the inhibition of Caspase-8 seems to have a more pronounced effect on the levels of apoptosis compared to other Caspase enzymes suggesting this caspase may play a more prominent role in the bacteria-induced cell death of the host animal. We will also ascertain whether this inhibitor can also have a similar effect under modeled microgravity.
Bibliography Type:	Description: (Last Updated: 04/20/2026)
Articles in Peer-reviewed Journals	Vroom MM, Rodriguez-Ocasio Y, Lynch JB, Ruby EG, Foster JS. "Modeled microgravity alters lipopolysaccharide and outer membrane vesicle production of the beneficial symbiont Vibrio fischeri." npj Microgravity. 2021 Mar 8;7(1):8. https://doi.org/10.1038/s41526-021-00138-8 ; PMID: 33686090; PMCID: PMC7940393 , Mar-2021